

Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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1 The CFA Franc Zone

The CFA franc zone is one of the longest-running fixed exchange rate arrangements in the global economy. It links a group of West and Central African countries to an external anchor currency through two monetary unions: the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CEMAC). The institutional architecture has historically rested on four central features: a fixed parity against the anchor currency, a convertibility guarantee by the French Treasury, free capital mobility, and the pooling of foreign exchange reserves ([Hadjimichael and Galy, 1997](#); [Banque de France, 2015](#)). These features have made the CFA franc zone an important case study in international macroeconomics because it combines monetary credibility and exchange-rate stability with a substantial loss of nominal exchange rate flexibility and monetary policy autonomy. The central policy question is whether this arrangement has supported or constrained economic performance.

2 Replication of the Original Study

Alassane Diallo and Adama Ba sought to explore the impact of the peg from the French franc to the euro on the economies of the countries in the CFA-zone ([Diallo and Ba, 2024](#)). The literature is divided about whether they have experienced a stronger or weaker growth after the peg shift, compared to what would have been without France joining the euro-zone. On one hand we could expect a high growth rate due to the access to an intensive trade market with the European Economic Community ([Hadjimichael and Galy, 1997](#); [Dearden, 1999](#)). But on the other hand there are many uncertainties linked to the consequence of being pegged to a strong currency like the euro, this could undermine expectations following the switch of anchor currencies ([Zafar, 2005](#)).

A French Senate report later made a similar point, stating that there was no general study allowing one to say clearly whether membership of the Franc Zone was an advantage or a handicap for member countries ([Goulet and Lurel, 2020](#)).

Therefore, the authors of the paper wanted to address this gap by estimating the impact of the euro changeover on CFA-zone economies. Their goal was first to allow for more empirical evidence to strengthen the debates but also to inform CFA-zone policy makers. For their analysis they chose to use GDP per capita as the outcome variable of interest.

Diallo and Ba use the synthetic control method to construct counterfactual GDP paths for each treated CFA country ([Abadie, 2021](#); [Abadie et al., 2010](#)). This is done through the construction of a weighted average of comparable non-CFA countries, based on pre-treatment characteristics. This method provides a transparent comparison group for our treated countries, and it can produce a more credible counterfactual than with a single control country.

The thirteen treated countries of the CFA-zone are divided between two economical communities: the West African Economic and Monetary Community (WAEMU) and the Central African Economic and Monetary Community (CAEMC), as shown in Table 1.

The donor countries all have fixed exchange rates regimes and have been matched based on other economic or institutional characteristics. The authors first included African countries with similar socioeconomic characteristics as the treated countries (Botswana, Eswatini, Lesotho, Mauritius, Morocco, Namibia and Seychelles) and they then added other countries with the similar level of development (Bangladesh, Barbados, Bhutan, Bolivia, Dominican Republic, Ecuador, Grenade, Laos, Oman, Panama, Saint Kitts & Nevis and Saint-Lucia). We have

Table 1: Treated countries and regional classification

Country	ISO3	Zone	Entry
Benin	BEN	WAEMU	Original CFA member
Burkina Faso	BFA	WAEMU	Original CFA member
Côte d’Ivoire	CIV	WAEMU	Original CFA member
Mali	MLI	WAEMU	Rejoined WAEMU in 1984
Niger	NER	WAEMU	Original CFA member
Senegal	SEN	WAEMU	Original CFA member
Togo	TGO	WAEMU	Original CFA member
Guinea-Bissau	GNB	WAEMU	Joined WAEMU in 1997
Cameroon	CMR	CEMAC	Original CFA member
Central African Republic	CAF	CEMAC	Original CFA member
Chad	TCD	CEMAC	Original CFA member
Republic of Congo	COG	CEMAC	Original CFA member
Gabon	GAB	CEMAC	Original CFA member
Equatorial Guinea	GNQ	CEMAC	Joined CEMAC in 1985

chosen to add three countries that are also similar to allow for more precise results (Turkmenistan, Guyana and Iraq).

The treatment date is 2002, as it was the date of the French Franc to Euro peg changeover. This means that the synthetic control weights will be computed on the similarities of the characteristics from 1980 to 2002.

The paper finds positive post-treatment output gaps for several countries, however those results are not statistically significant for any countries, except for Equatorial Guinea.

3 Replication Methodology and Data

The replication follows the original paper’s country-level synthetic control design. For each treated CFA franc country, we estimate a counterfactual GDP path using a weighted average of non-CFA donor countries, and then compare the observed post-2002 GDP path with this synthetic counterfactual. The treatment year, time period, and country sample all follow the original paper’s specifications, while variable selection and construction, and missing data approaches diverged. Further details on the variable construction, and missing-data treatment are reported in Appendix A: Table 3, and Table 4.

Let Y_{it} denote GDP per capita for country i in year t . For a treated country $i = 1$, the unobserved counterfactual outcome after treatment is approximated using a weighted average of J donor countries:

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad (1)$$

where the donor weights satisfy the standard synthetic control constraints:

$$w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1. \quad (2)$$

The weights are chosen to make the synthetic control resemble the treated country as closely as possible before treatment. Let X_1 denote the vector of pre-treatment characteristics for the treated country, and let X_0 denote the corresponding matrix for the donor countries. The donor-weight vector W is selected to minimise the distance between the treated country and the weighted donor average:

$$\min_W (X_1 - X_0 W)' V (X_1 - X_0 W), \quad (3)$$

subject to the constraints in Equation 2. The matrix V determines the relative importance assigned to each predictor in matching the treated country to the donor pool. In the reconstructed specification, the predictors

include agriculture, government share, investment share, official development assistance, foreign direct investment, labour, institutional quality, and selected pre-treatment GDP values. Industry is excluded from the final specification because of missingness and data-quality concerns, as discussed in Appendix B.

The estimated treatment effect is the post-treatment gap between the observed outcome and the synthetic counterfactual:

$$\hat{\alpha}_{1t} = Y_{1t} - \hat{Y}_{1t}^N = Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}, \quad t \geq 2002. \quad (4)$$

A positive value of $\hat{\alpha}_{1t}$ indicates that actual GDP per capita was above the synthetic counterfactual, while a negative value indicates that actual GDP per capita was below the counterfactual. The interpretation of these gaps depends on the quality of the pre-treatment fit: if the synthetic control does not reproduce the treated country’s pre-2002 path closely, the post-treatment gap is less informative.

The replication is best understood as a reconstructed replication rather than a mechanical reproduction of the original paper. Two issues were especially important. First, the GDP series used in the original paper could not be identified exactly from the published documentation. Second, the paper does not report how missing observations were handled, even though missing outcomes or predictors can affect donor eligibility, predictor averages, and estimated synthetic controls. These issues are discussed in Appendix B; the final variable definitions are summarised in Table 3, and the missing-data procedures are summarised in Table 4.

The analysis proceeds in four steps:

Step	Description
1	The country-year panel is reconstructed from public datasets.
2	Variables are harmonised and missing values are handled using documented rules.
3	One synthetic control is estimated for each treated country, producing donor weights, predictor-balance statistics, actual-versus-synthetic GDP paths, and post-treatment gaps.
4	In-space placebo tests are used to compare the treated-country gaps with gaps obtained when the same procedure is applied to donor countries as placebo-treated units.

Several additional reporting issues in the original paper also shaped the replication strategy. These include donor-pool ambiguity, inconsistent country naming, unclear variable definitions, inflation-source inconsistencies, table-label errors, and numerical values that are difficult to reconcile with the stated data structure. These issues do not invalidate the paper’s research question, but they reduce confidence that the published tables provide sufficient information for an exact replication. The purpose of the replication is therefore to evaluate the robustness of the paper’s broad conclusions using a transparent, reproducible data construction and estimation procedure.

4 Replication Results

The country weights computed by the synthetic control method, which then allows us to create the synthetic GDP per capita, are shown in Table 4. We were not able to replicate the original values exactly as the GDP data were not the same. However, as in the original paper, the synthetic control analysis relies heavily on Bangladesh and Laos. The first has a positive weight for 9 out of the 13 treated countries, being especially present for Mali as well as Burkina Faso. Laos has positive weights for 10 out of the 13 countries, an explanation for this is that the country has similar pre-treatment income to a lot of the treated countries compared to the other donor countries. Iraq, one of the countries that we added, had positive weights for 9 out of 13 treated countries.

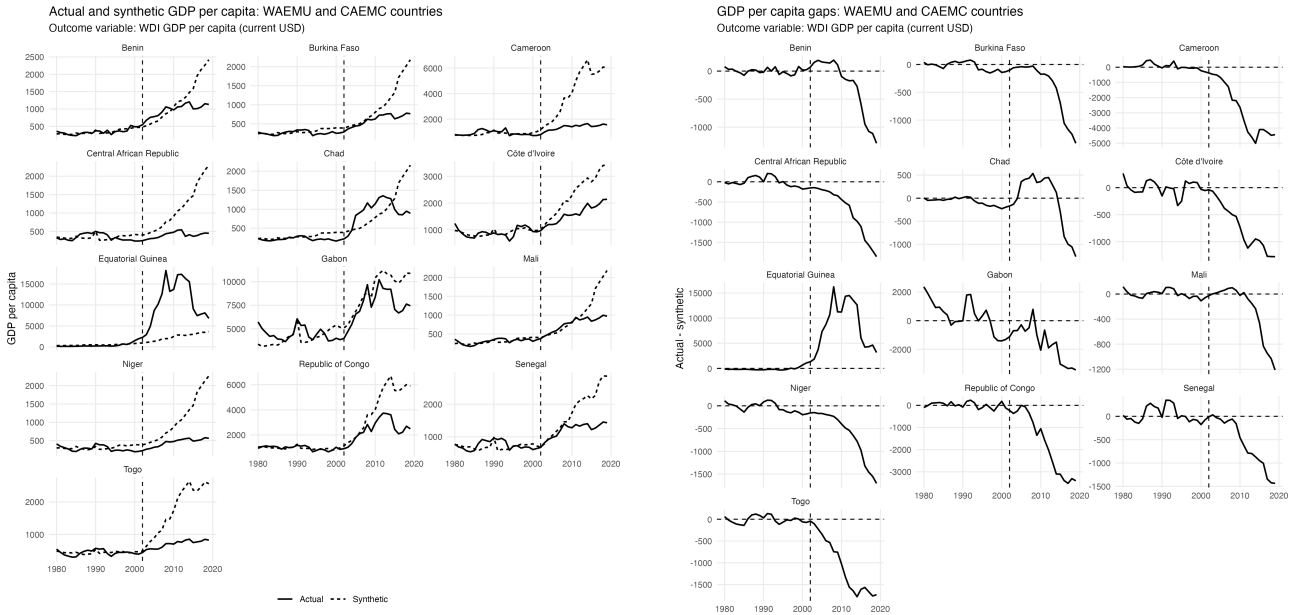
Figure 1a presents the result of our analysis using the synthetic control method, with the comparison of the synthetic control GDP (dotted line) per capita and the actual GDP per capita (solid line) for each treated country. The vertical dotted line marks the year 2002, our treatment, and thus before it both curves should be similar to attest to the good fit of our method. Thus the closer the dashed line is to the full line during the 1980 to 2002 pre-treatment period, the better the synthetic control matches the treated country. And even

Table 2: Estimated synthetic control weights by treated country

Donor country	Benin	B. Faso	Côte d'Ivoire	Mali	Niger	Senegal	Togo	Cameroon	CAR	Chad	RC	Gabon	Eq. G.
Bangladesh	.771	.901	.353		.946	.757	.356	0	.072	.785	.934	0	0
Barbados	.002	0	0		0	0	0	0	0	0	0	.345	0
Bhutan	.124	0	0		0	0	0	0	.053	0	0	0	.988
Bolivia	0	0	.093		0	0	.261	.011	.002	0	0	.075	0
Botswana	0	0	0		0	0	0	0	0	0	0	.007	0
Cabo Verde	0	0	0		0	0	0	0	0	0	0	0	0
Dominica	0	0	0		0	0	0	0	0	0	0	0	.012
Ecuador	.001	0	.110		0	0	0	0	.013	0	0	.159	0
Eswatini	0	0	0		.008	0	0	0	0	0	0	0	0
Grenada	0	0	0		0	0	0	0	0	0	0	0	0
Lao PDR	.094	.097	.120		.040	.236	.047	.301	.017	.191	.066	0	0
Lesotho	0	0	0		0	0	.203	.486	0	0	0	0	0
Mauritius	0	0	0		0	0	0	.083	0	0	0	0	0
Morocco	0	0	.225		0	0	0	0	0	0	0	0	0
Namibia	0	0	.074		0	0	.090	0	0	0	0	.407	0
St. Kitts and Nevis	0	0	0		0	0	0	0	0	0	0	0	0
St. Lucia	0	0	0		0	0	0	0	0	0	0	0	0
Seychelles	0	0	0		0	0	0	0	0	0	0	0	0
Turkmenistan	0	0	0		0	0	0	.187	.738	0	0	.725	0
Guyana	0	0	0		0	0	0	0	0	0	0	0	0
Iraq	.008	.002	.026		.006	.008	.043	.015	.022	.023	0	.033	.249

though we can note some imperfection, especially when the growth is highly variable, the synthetic and treated pathways are very similar for most countries.

Figure 1b presents the difference between the synthetic and actual GDP per capita. We can note again the good fit of our analysis as in the 1980-2002 period the gaps are generally near 0. This suggests that we have credible counterfactuals that allow us to address the euro-peg period.



(a) Actual and synthetic GDP per capita for treated CFA countries

(b) GDP per capita gaps between actual and synthetic controls for treated CFA countries

Figure 1: Synthetic control method outputs

Both figures are tools to compare the growth that actually happened with the treatment and our proxy growth. In our result, we found that almost all countries showed increasingly large negative GDP per capita gaps as the actual output fell below our counterfactual, with only Equatorial Guinea having a positive output gap. This suggests that the changeover to a euro peg had negative effects on the countries of the CFA-zone. Our results can be explained by an overvaluation of their currency because of the peg to a strong euro. This would make trade more difficult for our treated countries. However we cannot draw those conclusions using only the GDP per capita as it has multiple different input, it would thus be interesting to overview another output that is more closely related to trade, for example imports and exports.

However our results conflict with the one of the original paper. Indeed, Diallo & Ba (2024) found a positive output gap for multiple countries, for instance Chad, the Central African Republic or Mali. Thus their result would suggest that the peg of CFA Franc to the euro may boost the real GDP per capita compared to the

synthetic output. Due to the data and methodology issues in the original paper, we cannot be sure about the exact reasons for this discrepancy, but it is likely that the different GDP series, missing-data treatments, or errors in the original implementation played a major role.

4.1 Replication placebo testing

Figure 2 presents the results of our placebo significance testing. We followed the in-place placebo method, which consists of analysing each donor country as if it had received the treatment. Then we have estimated the synthetic control for each donor country and we can then compute the output gap for each country. The aim of this approach is to compare the difference between the pre-treatment and post-treatment output gap of each treated country with the analogous output gaps of the donor countries.

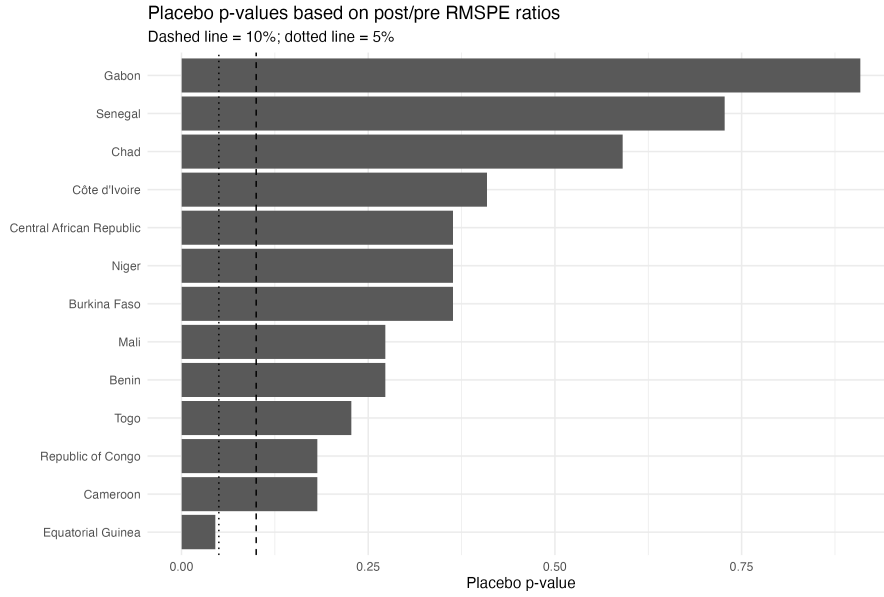


Figure 2: Placebo p-values for the synthetic control method

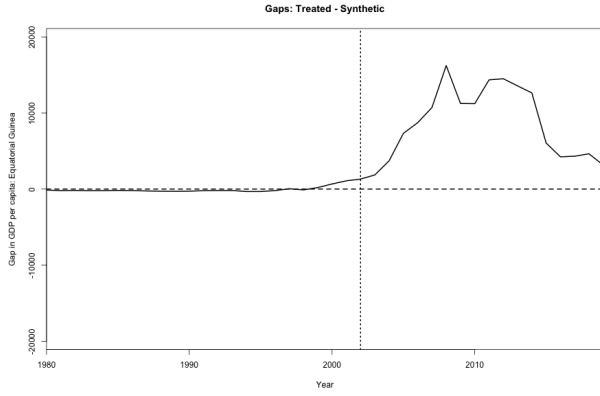
The p-values are computed using the post/pre Root Mean Square Per Error ratios. The ratio is computed for all countries, and we then rank the results of the donor countries. Then we can observe how unusual, relatively, the ratio of each country is by seeing how low or high in the ranking it is, with the share of countries with a more extreme ratio determining the placebo p-values.

With this method, we obtained the same results as the authors of the original paper. At the 5 and 10% level, we find that only the result of Equatorial Guinea was statistically significant, while all the other treated countries do not have any significant results.

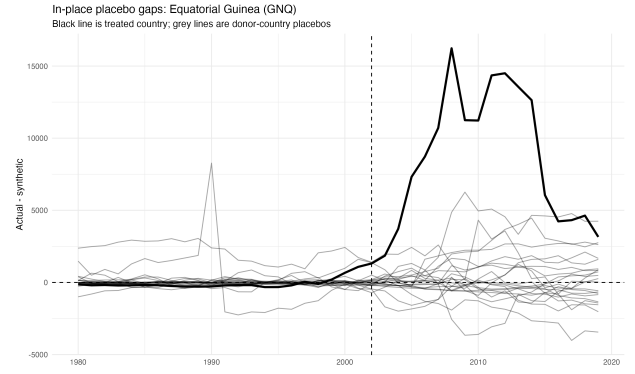
4.1.1 Equatorial Guinea

Equatorial Guinea is an unusual case. In Figure 3a we can observe the result for its synthetic control, with the output gap. And indeed we observe a consistently positive GDP per capita gap. We can also note that those results are the one that match the most those of the original paper, in terms of closeness of the numbers of the output gap. In Figure 3b, we can observe graphically that those results are significant.

Equatorial Guinea should be interpreted cautiously because its post-treatment GDP path coincides with the oil boom. Large oil discoveries in the 1990s transformed the economy, and by the early 2000s oil and gas had become the major driver of exports and government revenue (Tordo et al., 2011; Vines, 2009). The authors of the original paper also reference studies that show an undervaluation of its exchange rates (Feindouno et al., 2020). As such, Equatorial Guinea may not be a typical case for the CFA franc zone.



(a) Equatorial Guinea's GDP per capita gap between actual and synthetic control



(b) Placebo gaps for Equatorial Guinea compared to donor countries

Figure 3: Results for Equatorial Guinea

5 Extension: Trade

5.1 Extension Motivation

The GDP replication suggests that the shift to the euro anchor did not generate a clear increase in output across the CFA franc zone. However, the peg may still have impacts on trade outcomes, and the direction of these effects is not agreed on.

The impact of the euro peg on the trade channel is theoretically ambiguous. On the one side, the euro peg could strengthen trade with Europe by reducing exchange-rate uncertainty, lowering transaction costs, and increasing monetary credibility. This mechanism is consistent with broader currency-union literature, which finds that common currencies and reduced exchange-rate volatility can increase bilateral trade (Rose, 2000; Glick and Rose, 2016). In the specific CFA context, Martínez-Zarzoso (2019) finds positive sectoral trade effects from the euro for some CFA exports to euro-area countries.

On the other side, because the CFA franc moves with the euro, euro appreciation can appreciate the CFA franc relative to non-euro currencies. This may weaken export competitiveness and make euro-area imports relatively more expensive, encouraging firms and consumers to shift towards non-euro suppliers. This concern is supported by Coulibaly (2014), who argues that after the switch to the euro, CFA-zone competitiveness became more dependent on movements in the anchor currency.

The extension therefore asks whether there is evidence that the euro peg increased trade with the euro area, or reduced it, to add empirical evidence to the discussions on the direction of the impact on euro trade.

5.2 Extension Methodology

Focusing on the trade channel, we considered different variables. The ones that caught our attention are the share of imports coming from the Euro Area (MEU) and the share of exports going to the Euro Area (XEU). These are the most relevant variables in our context because they allow us to partly avoid a confounding issue with the 1994 devaluation of the CFA franc. The CFA franc was devalued by 50 percent in January 1994, partly to restore external competitiveness (International Monetary Fund, 1994a,b). This makes level variables like exports and imports as a share of GDP, as shown in Figure 4, unsuitable as SCM outcomes. Using trade shares reduces, but does not eliminate, the risk that the 1994 devaluation mechanically drives the results. As a result, these variables are better suited to asking whether the euro peg changed the composition of trade toward or away from Europe.

The subsequent step is to apply the SCM to our two extension variables, MEU and XEU, following the same four-step structure as the replication: pre-treatment fit, actual versus synthetic paths, gap analysis, and placebo inference.

As previously stated, the quality of the pre-treatment fit between the synthetic counterfactual and the treated unit is central to the credibility of the synthetic control method. This can be assessed by examining Figure 5a, which plots the actual EU trade shares of treated countries alongside their synthetic counterfactuals. For the import variable MEU, the fit is reasonable across most countries over the 1980 to 2002 period. However there

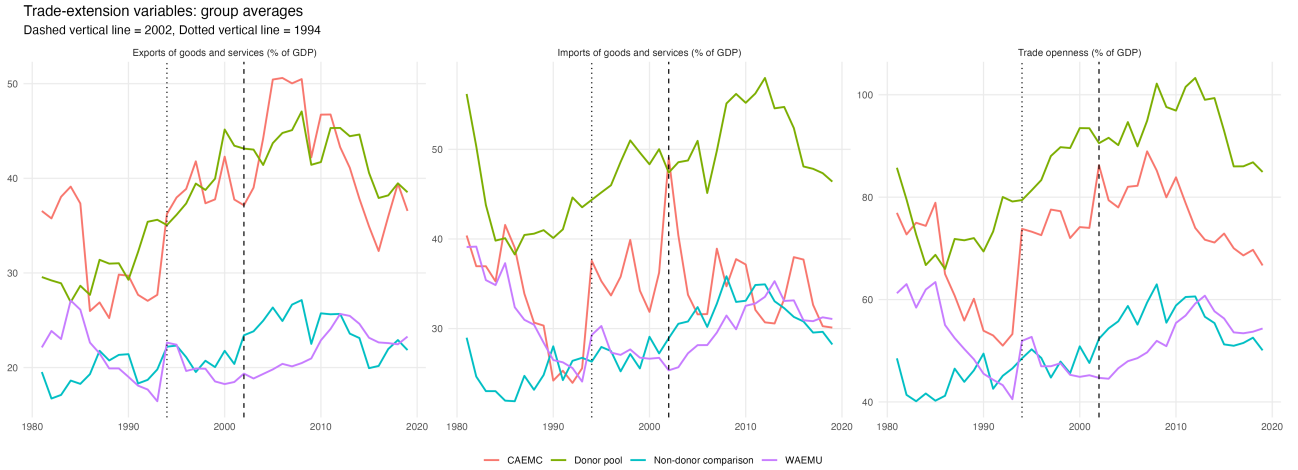
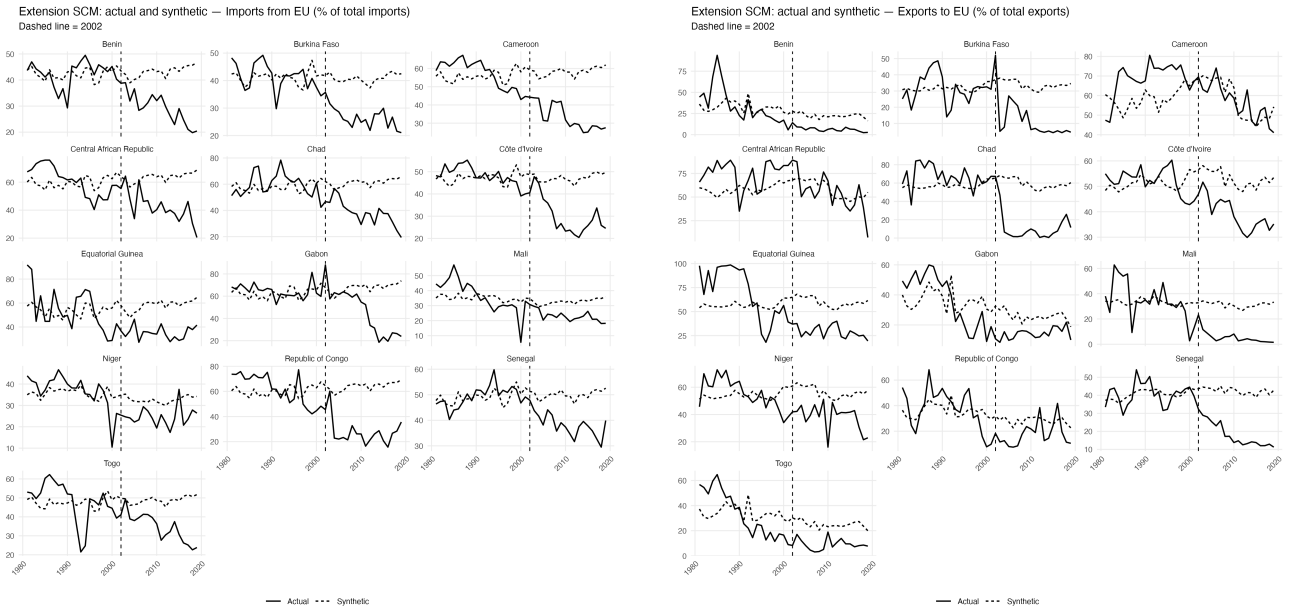


Figure 4: Level trade variables for different groups of countries

are still some imperfections remaining, as trade shares tend to be more volatile than GDP per capita, making them inherently harder to match precisely. The pre-treatment fit for MEU is therefore weaker than the one obtained in the GDP replication. For the export variable XEU, the pre-treatment fit is more uneven still, with several countries showing larger discrepancies before 2002.



(a) MEU paths between actual and synthetic controls for treated CFA countries

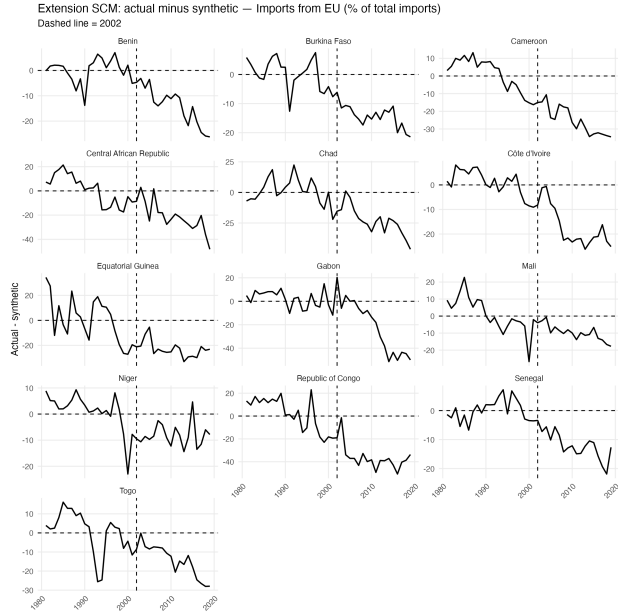
(b) XEU paths between actual and synthetic controls for treated CFA countries

Figure 5: Trade extension paths between actual and synthetic controls for treated CFA countries

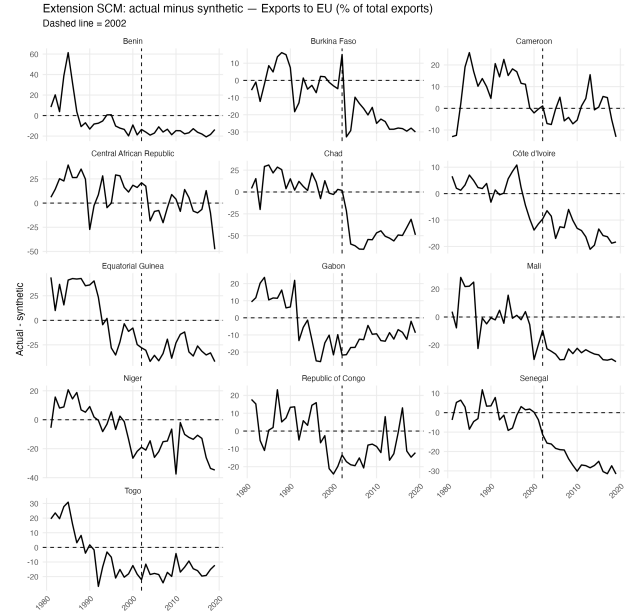
Then Figure 6 presents the gap between the actual and synthetic trade shares with Europe for each country. This provides another way of assessing pre-treatment fit, as the gaps should oscillate around zero before 2002. For the trade shares this is less clean than in the GDP replication, but the gaps remain broadly centred around zero for most countries, supporting the use of these synthetic controls as credible counterfactuals.

On the whole, while the pre-treatment fit for both MEU and XEU is weaker than the one obtained in the GDP replication, it remains sufficiently close to zero to justify proceeding with the post-treatment analysis.

Focusing now on post-treatment gaps, the results diverge between the two variables, MEU and XEU. For MEU : we observe only negative gaps across all countries, meaning the share of imports coming from the EU fell following the pegging to the euro. For XEU, no clear pattern emerges from the gaps, with no consistent directional trend across treated countries



(a) MEU gaps between actual and synthetic controls for treated CFA countries

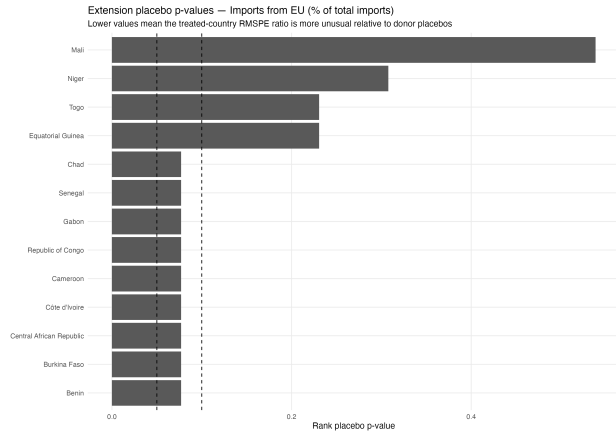


(b) XEU gaps between actual and synthetic controls for treated CFA countries

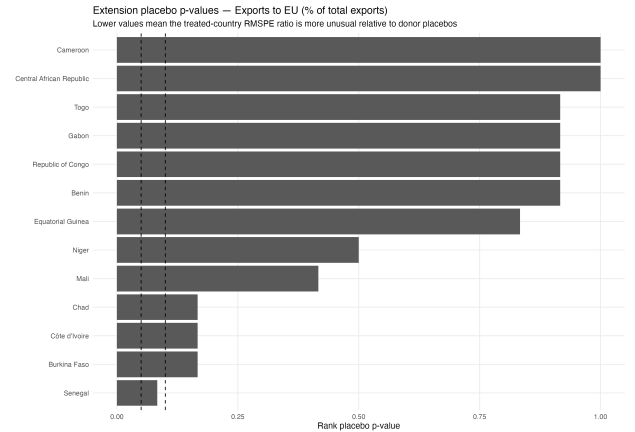
Figure 6: Trade extension gaps between actual and synthetic controls for treated CFA countries

5.3 Extension placebo analysis

To assess whether these post-treatment gaps are statistically unusual relative to what could occur by chance, we turn to placebo inference, shown in Figures 7a and 7b. Full visualisations of the placebo results are included in Appendix C. One important constraint applies to both variables: with only 13 donor countries, the reference distribution contains 14 units in total, meaning the minimum achievable rank p-value is 1/14, approximately 0.077. Significance at the 5% level is therefore unreachable by construction for either variable.



(a) Placebo p-values for MEU



(b) Placebo p-values for XEU

Figure 7: Placebo p-values for trade extension

Within that constraint, results diverge depending on whether we look at the import or export side. For the share of imports coming from the EU, 9 out of 13 countries show significant negative gaps, by reaching the lowest attainable p-value. This indicates that EU import shares fell below their synthetic counterfactual following the change in peg.

In contrast, for the share of exports going to the EU, the evidence of any effect is considerably weaker. Almost no treated countries display significant gaps relative to the placebo distribution, with Senegal representing the only borderline exception at the 10% significance level. The euro peg therefore appears to have had little detectable effect on the share of exports directed toward the EU.

Overall, the significant tests using the placebo method confirm that the negative gaps in the import shares are statistically significant for a majority of treated countries, but as expected, the export shares do not yield significant results to interpret.

5.4 Interpretation of extension placebo testing results

The placebo results therefore point to an asymmetric trade response: the euro peg is associated with a statistically detectable fall in EU import shares for many CFA countries, but not with a comparable change in EU export shares. The results suggest a compositional adjustment in import sourcing, where CFA countries appear to have moved away from EU suppliers relative to their synthetic counterfactuals.

These results are consistent with the overvaluation channel interpretation. The euro-dollar chart in Figure 8 is useful here. It shows a sharp appreciation of the euro after the early 2000s, especially between 2003 and 2008. Since a large share of international trade is invoiced in dominant currencies, especially the US dollar, exchange-rate movements against the dollar can strongly affect import prices and trade values (Gopinath et al., 2020; Boz et al., 2022). This appreciation would therefore have raised the relative price of euro-area imports for CFA countries compared with imports from dollar-linked or non-euro suppliers.



Figure 8: Euro-dollar exchange rate, 1999-2026

5.4.1 CFA franc overvaluation channel: Cotton

There is literature noting this effect for some specific goods, including cotton. Pigeaud and Sylla (2021) discuss cotton as a concrete example, arguing that CFA exporters can be squeezed when export prices are denominated in dollars, but domestic costs and producer prices are denominated in the CFA franc. They note that this was particularly important for Burkina Faso, Mali, and Benin, where cotton is a major export, and describe the 2004–2007 euro appreciation as damaging for Burkina Faso’s cotton sector when combined with falling world cotton prices.

This interpretation is supported by Bellocq and Silve (2007), who conducted an AFD study of the Burkina Faso cotton crisis, which documents the losses of cotton companies and the fiscal support required from the Burkinabe state during this period. It is also consistent with the World Bank’s Burkina Faso diagnostic, which states that the fall in the CFA-franc price of cotton was aggravated by the rise of the euro, and that euro appreciation created major competitiveness problems for exports, especially cotton. Earlier policy analysis for the European Parliament made the same point during the early euro period, arguing that CFA cotton producers’ competitiveness declined as the euro strengthened against the dollar.

More broadly, IMF work on WAEMU trade highlights the macroeconomic importance of cotton exports for Benin, Burkina Faso, Mali, and Togo, while studies of CFA real exchange-rate misalignment argue that appreciation of the CFA franc against the dollar can reduce competitiveness outside the euro area. These sources support the idea that export effects may be sector-specific, which might suggest why aggregate EU export shares may not show a strong average response in the analysis results.

The extension results partly contrast with literature that emphasises positive trade effects from fixed exchange rates or currency unions. Martínez-Zarzoso finds that the euro had positive and significant effects for some CFA export flows to euro-area countries, although the effects vary by sector and direction of trade. Our results are not necessarily inconsistent with this sectoral evidence, because the extension uses aggregate EU trade shares rather than bilateral sector-level flows. Positive effects in specific export sectors may be offset by negative

effects in others, or may be too small to change aggregate export shares. The absence of strong export-share effects may also reflect the structure of CFA exports: many are primary commodities whose destinations are impacted by global demand, resource endowments, long-term contracts, and commodity-market conditions, as well as exchange-rate impacts.

There are also methodological limitations with the extension analysis. The extension uses a synthetic control framework adapted from the GDP replication, but the best predictors of trade composition may differ from the best predictors of GDP per capita. Variables such as bilateral distance, port access, logistics quality, product composition, tariff exposure, colonial trade networks, and partner-country demand may be more relevant for predicting EU trade shares than the macroeconomic predictors available in the reconstructed GDP panel. These limitations are consistent with the gravity-trade literature, where bilateral trade is typically modelled as depending on economic mass, distance, trade costs, borders, policy barriers, and multilateral resistance terms (Anderson and van Wincoop, 2003; Head and Mayer, 2014).

The donor pool is also limited by missing trade data, leaving a smaller set of usable placebo countries (13 of 20). For this reason, care should be taken when interpreting the result of no statistically significant effects for the export sector. Export-side effects may be weaker, more sector-specific, or simply harder to detect using aggregate trade-share outcomes.

6 Conclusions

This project began as a replication exercise for the work of Diallo and Ba, asking whether the CFA franc's shift from the French franc anchor to the euro anchor changed economic outcomes in the CFA franc zone. However, the project became a reconstruction exercise instead, because the original paper contains data gaps, methodological ambiguities, and reporting errors that prevent a fully mechanical replication. In particular, the GDP variable could not be identified exactly, the treatment of missing values is not documented, and several tables contain inconsistencies. The analysis therefore reconstructs the study using public data, explicit variable choices, and documented cleaning decisions.

The GDP results support a cautious version of the original paper's conclusion. The reconstructed SCM estimates do not show clear evidence of broad GDP gains after 2002. Most countries have negative average post-treatment GDP gaps, although these are generally not statistically unusual in the placebo analysis. Equatorial Guinea remains the main positive exception, but its oil dependence and unusual GDP trajectory make it difficult to interpret as evidence of a general CFA-zone effect.

The trade extension gives a more specific picture. Although aggregate GDP effects are weak, the euro peg appears to have affected trade composition, especially import sourcing. The strongest extension result is that EU import shares fall relative to synthetic controls for many CFA countries. This suggests that the EU became less important in import baskets than it would have been under the counterfactual, rather than that total imports necessarily fell.

This finding supports the overvaluation and relative-price narrative in the CFA franc literature. Because the CFA franc moves with the euro, euro appreciation can make euro-area goods relatively more expensive and can reduce the CFA value of dollar-denominated export revenues. This is consistent with concerns about competitiveness pressures under the euro peg, including those discussed for cotton-exporting countries. At the same time, the results do not strongly support the alternative narrative that the euro anchor broadly increased Europe-oriented trade through credibility, lower transaction costs, or reduced exchange-rate uncertainty.

The main limitation of the extension is that the SCM specification was adapted from the GDP replication, while trade composition may require different predictors, such as distance, logistics, sectoral export structure, tariffs, commodity dependence, and partner-country demand. The donor pool is also limited by missing trade data, reducing the number of placebo countries. Future work could therefore use richer bilateral and sectoral trade data, expand the donor pool, and test whether effects differ across major export sectors such as cotton, oil, cocoa, and minerals.

Overall, the results suggest that the euro peg did not generate broad GDP gains across the CFA franc zone, but may have mattered through trade composition. A plausible conclusion is that the euro anchor provided monetary continuity while also reshaping external adjustment pressures, especially through relative prices and import sourcing.

A Data Construction and Variable Definitions

The reconstructed dataset combines World Development Indicators, Penn World Table, Polity5, FAOSTAT, and IMF Direction of Trade Statistics ([World Bank, 2026](#); [Feenstra et al., 2015](#); [Marshall and Gurr, 2020](#); [Food and Agriculture Organization of the United Nations, 2026](#); [International Monetary Fund, 2026](#)).

Table 3: Variable construction in the original paper, updated replication, and trade extension

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Real GDP per capita / selected GDP per capita	<code>gdp_selected;</code> <code>wdi_gdp_pc_current</code>	Main GDP outcome in replication	World Development Indicators	The original paper studies real GDP per capita. In the replication, alternative GDP measures were compared and WDI GDP per capita, current US dollars, was selected as the closest empirical match to the paper’s reported results.	Main divergence from the original paper; The exact GDP series in the paper could not be reproduced unambiguously, considering numerous WDI and PWT indicators.
Agriculture value added	<code>agriculture</code>	SCM predictor	WDI and FAOSTAT	Agriculture, forestry, and fishing, value added as a percentage of GDP. WDI indicator: <code>NV.AGR.TOTL.ZS</code> . Alternative agriculture series were used to fill major gaps.	Retained in the final predictor set.
Industry value added	<code>industry</code>	Original/intended SCM predictor	WDI	Industry including construction, value added as a percentage of GDP. Main WDI indicator: <code>NV.IND.TOTL.ZS</code> .	Included in the original paper’s predictor set but removed from the final replication because of missingness and data-quality problems.
Government expenditure share	<code>govt_share</code>	SCM predictor	Penn World Table	Government consumption share from PWT, using <code>csh_g</code> .	Corresponds to the government expenditure control described in the original paper.
Investment share / gross fixed capital formation	<code>invest_share</code>	SCM predictor	Penn World Table	Investment or capital-formation share from PWT, using <code>csh_i</code> .	Corresponds to the paper’s gross fixed capital formation control.
Official development assistance	<code>oda_share</code>	SCM predictor	World Development Indicators	Main WDI series used include net ODA received. <code>DT.ODA.ODAT.GD.ZS</code> .	ODA as a share of national income. Retained in the final predictor set.

Variable	Code name	Role in analysis	Source in updated replication	Construction / definition	Notes
Foreign direct investment	fdi	SCM predictor	World Development Indicators	Foreign direct investment, net inflows as a percentage of GDP. WDI indicator: BX.KLT.DINV.WD.GD.ZS .	Retained in the final predictor set.
Labour / employment	labour	SCM predictor	Penn World Table	Employment as a share of population, constructed as $100 * \text{emp} / \text{pop}$.	Corresponds to the original paper's labour factor, described as the proportion of the population in work.
Political regime / institutions	polity2	SCM predictor	Polity5	Polity2 institutional score, ranging from autocracy to democracy (-10 to +10).	Interpolation and extrapolation were applied where needed to obtain usable pre-treatment panels.
1994 CFA devaluation dummy	devaluation_1994	Original control / historical adjustment	Constructed	Dummy variable intended to capture the January 1994 CFA franc devaluation.	The original paper mentions this control. In the replication code, it was not retained as a standard SCM predictor because it has no useful cross-donor variation.
Trade openness	trade_openness	Trade-extension outcome	World Development Indicators	Exports plus imports of goods and services as a percentage of GDP. WDI indicator: NE.TRD.GNFS.ZS .	Used to test whether the euro peg changed aggregate trade exposure.
Exports of goods and services	exports_gdp	Trade-extension outcome	World Development Indicators	Exports of goods and services as a percentage of GDP. WDI indicator: NE.EXP.GNFS.ZS .	Used to examine export intensity.
Imports of goods and services	imports_gdp	Trade-extension outcome	World Development Indicators	Imports of goods and services as a percentage of GDP. WDI indicator: NE.IMP.GNFS.ZS .	Used to examine import intensity.
Exports to EU share	XEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times exports to the EU divided by exports to the world.	Measures export composition toward the EU.
Imports from EU share	MEU	Trade-extension outcome	IMF Direction of Trade Statistics	Computed as 100 times imports from the EU divided by imports from the world.	Measures import composition toward the EU.

Table 4: Missing-data treatment in the reconstructed panel

Variable or issue	Method used	Notes
Polity score	Nearest-observation fill	Applied generally to fill missing institutional observations.
Agriculture	FAOSTAT supplement; nearest-observation fill	FAOSTAT was used to fill missing WDI agriculture values where possible. Remaining gaps were handled through the imputation plan.
Industry	Nearest-observation fill in data construction; excluded from final SCM	Although industry was included in the broader data construction process, it was dropped from final SCM because data quality remained problematic.
ODA	Country-specific and general nearest-observation fill	Several donor countries required endpoint filling. ODA construction remains a reconstruction choice because the original paper does not clearly document the denominator.
FDI	Zero-fill for selected early gaps; nearest-observation fill generally	Zero-filling was used for selected cases where missing early values were treated as approximately zero; other gaps were filled using nearest observations.
Labour	Country-specific and general nearest-observation fill	Used for countries with early, late, or broad labour-data gaps.
Government share	Nearest-observation fill	Constructed from PWT and filled where necessary.
Investment share	Nearest-observation fill	Constructed from PWT and filled where necessary.
GDP variables	Nearest-observation fill; selected linear interpolation	Several GDP alternatives were constructed and filled during data preparation, though the final outcome was WDI GDP per capita in current US dollars.
Donor eligibility	Donor screening before SCM	Donors entirely missing at least one active predictor over the pre-treatment period were excluded from the usable SCM donor pool.

B Issues in the Original Paper and Implications for Replication

B.1 Primary issues

A major obstacle in the replication was identifying the GDP variable used in the original paper. Although the paper describes the outcomes as real GDP per capita, the reported values could not be matched to any standard GDP series available in either WDI or Penn World Table. We tested the full set of WDI GDP per capita measures, PWT output- and expenditure-side GDP variables, and constructed alternatives combining WDI aggregate GDP measures with PWT population to generate per-capita series. None of these reproduced the GDP values reported in the original tables. This created a fundamental reproducibility problem, because GDP per capita is both the main outcome and a key lagged predictor in the SC specification. We therefore treated the GDP series as an empirical reconstruction choice, rather than a directly observed replication input, selecting WDI GDP per capita in current US dollars, because it provided the closest correspondence to the published synthetic control patterns.

This is important because synthetic control estimation requires a usable pre-treatment panel: missing values in the outcome or predictors can change which countries are eligible donors, alter predictor averages, and therefore affect the estimated synthetic counterfactual. In reconstructing the GDP analysis, missingness was substantial for several variables used or discussed in the original specification, especially sectoral predictors and some macro controls. The paper does not explain whether missing observations were dropped, interpolated, extrapolated, filled from alternative sources, or otherwise adjusted. We therefore had to make these choices explicitly. The final replication keeps both unimputed and imputed panels, fills selected gaps using documented rules, supplements agriculture with an additional source where possible, and drops industry from the final SCM specification because retaining it created serious data-availability and quality problems.

These two primary issues have major implications for the methods and results of this analysis, and also have many possible approaches. As such, we did not expect to be able to exactly match the results of the paper, but rather aimed to produce a transparent reconstruction of the analysis, with clearly documented choices.

B.2 Secondary Issues

In addition to the missing methodology for missing data, as well as the unclear source reporting and unidentifiable GDP data, there are a number of smaller issues identified in the paper. Some of these issues have moderate implications for the results, while some should have no major impact. However, the number of directly observable mistakes in the paper substantially reduces confidence in the methods and underlying work that are not described or explicitly shown.

Inflation source

The original paper describes the inflation data coming from WDI, however the published table values match the Penn World Table deflator-based series. This is unexpected, given the paper’s text explicitly frames the inflation comparison as being calculated from WDI values. This issues raised questions about the sources of other variables, when replication with the listed sources was not possible.

Donor-country naming: Dominica vs Dominican Republic

The original paper’s donor-pool description refers to the “Dominican Rep.”, while the synthetic control weights table lists “Dominica”. There are different countries, with different ISO codes, economic countries, and data histories. In the replication, we chose to align with the donor countries listed in the table and use Dominica.

Donor-pool size

The paper states that each treated country is matched using 21 control countries, but the donor-weight table only lists 20 countries. This brings some uncertainty about the exact donor countries used in the published analysis.

DRC label in CFA-zone tables

One of the published predictor-balance tables labels a treated country as “DRC”. The Democratic Republic of Congo is not a CFA franc-zone country and is not part of WAEMU or CEMAC. This is likely a mislabelling of the Republic of Congo, but it is also perhaps an indication that the wrong country was used as a treated country in the analysis. The replication uses explicit ISO3 country codes throughout to avoid ambiguity.

Devaluation dummy

The paper reports a 1994 CFA devaluation dummy as part of the predictor balance, but the values do not appear consistent with a dummy variable averaged over the stated pre-treatment period. Additionally, Senegal is reported as zero despite being a CFA country in 1994. For this reason, the devaluation dummy is not retained as a final SCM predictor in the replication.

Repeated GDP values in selected years

The GDP values reported in Appendix 4 of the paper repeat exactly across different years for the same country. This is clearly a mistake in the table, but means that one of the two columns is missing. This error was made more challenging by the inability to find their reported numbers through any calculation method with their listed sources.

Investment variable definition

The paper’s variable appendix appears to describe investment as a household-consumption share. This is inconsistent with the usual interpretation of investment of gross fixed capital formation. Since investment is an important growth predictor, this creates uncertainty over the exact variable used in the analysis: whether an incorrect variable was used, or whether the correct variables was used but not correctly reported.

Sectoral variable sources

The paper lists agriculture and industry as coming from WDI and Penn World Table, but Penn World Table does not provide the same standard sectoral value-added shares used in WDI. This makes the source documentation ambiguous. In the replication, agriculture is primarily taken from WDI and supplemented with FAOSTAT where needed, while industry is dropped because of missingness and data-quality problems.

OLS appendix observation counts

The paper’s OLS appendix reports 751 observations for country-specific regressions over 1990-2019. For 13 countries over 29 years, we would expect 377 observations. It is unclear what the 751 observations are, or how this table was generated, but this value suggest some error in calculation or duplication.

FDI copy error in OLS table

The paper’s OLS appendix has the reported FDI coefficient for Senegal as -43.7, vastly out of scale with the reported FDI coefficients for every other country (all between 15816 and 17245). Additionally, this value and its standard error match exactly the value for the ODA coefficient, suggesting there are errors in the table generation, possibly mistakenly copying the wrong value in.

Post-treatment RMSPE wording

The methodology section states that weights are chosen to minimise post-treatment RMSPE, but standard synthetic control methods chose weights using pre-treatment information only. Using post-treatment outcomes to choose weights would violate the logic of the method. This is likely a wording error, but would substantially change the method and results if it were the case.

Inflation table period label

The first column of the WAEMU inflation table appears to be incorrectly labelled. The table header indicates 1980-2021, while the other tables and surrounding text reference the pre-euro and post-euro period. These numbers do not match those identified by our replication for either the pre-euro, or total periods. Our ability to match the other inflation table values, except for the WAEMU section, paired with the incorrect label, suggests there may have been a mistake in the table creation or value calculations.

Zero inflation value for Sao Tome and Principe

The non-CFA inflation table reports Sao Tome and Principe as having exactly zero average inflation over the pre-period. This reflects missing data treatment rather than an actual inflation average, and the replication exercise finds a positive value here (31.7%) along with alignment with the rest of the table. This is one instance where missing value treatment is not explained but may have impacts on the results of the paper.

Synthetic control weights reported as percentage points

The paper describes the synthetic-control weights as being expressed in percentage points, but the values in the table are decimal weights. This is just a terminology issues, but matters for the interpretation of the table.

C Extension results: additional figures and tables

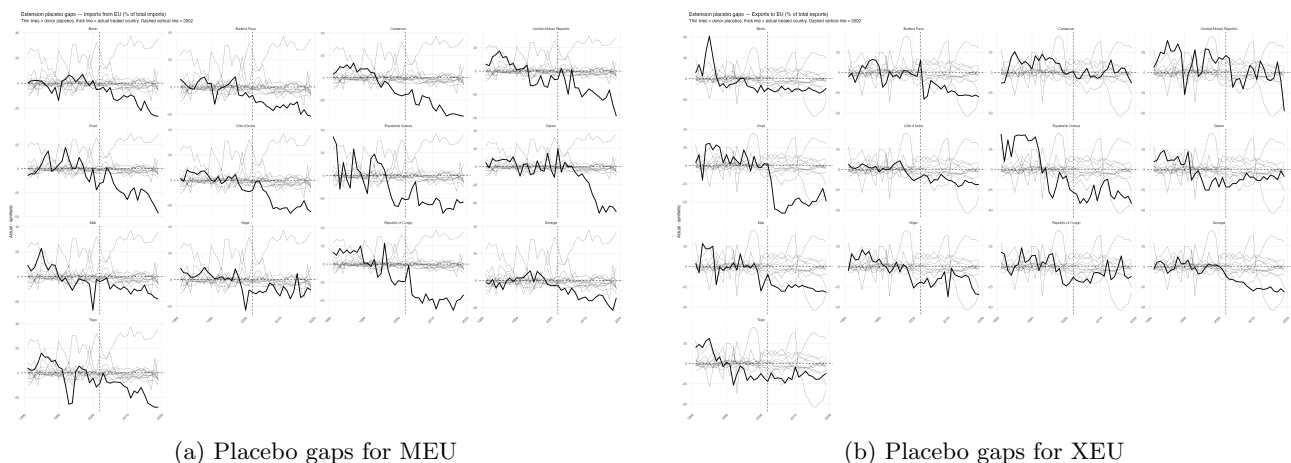


Figure 9: Extension placebo gaps

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